QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

IV SEMESTER

Minor - MG4DSCPHY202 - Basic Electronics and Electricity

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1.	Express current in terms of electron drift velocity.	[K][CO1]
2.	How do the temperature affect resistance of a material?	[U][CO1]
3.	Why are thermistors not suitable for very high-temperature measurements?	[U][CO1]
4.	Why no emf is induced when a conductor moves parallel to a magnetic field?	[U][CO1]
5.	What is meant by inductive reactance? State how it depends on frequency.	[U][CO2]
6.	Explain why a depletion layer forms at a PN junction.	[U][CO3]
7.	A student accidentally connects a PN junction diode in reverse bias to a high-voltage source. What is the likely outcome?	[A][CO3]
8.	An AC input of 12 V rms is applied to a full-wave rectifier. Calculate the peak voltage of the rectified output	[A][CO3]
9.	A capacitor filter fails to smooth the output at high load current. Analyse the reason.	[An][CO3]
10.	If emitter current of a transistor is 5 mA and collector current is 4.95 mA, find the base current.	[A][CO4]
11.	State the biasing condition for a transistor operating in CE configuration.	[U][CO4]
12.	In a CE amplifier, the base voltage is 2 V and emitter voltage is 1.3 V. Find V_{BE}.	[A][CO4]
13.	Give the symbol of an OP-Amp and outline the functions of each terminal.	[U][CO5]
14.	Why it is important to have high CMRR in practical OP-Amp applications?	[An][CO5]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Describe the factors that affect the magnitude of emf induced in a rotating coil. [U][CO1]

16. An AC source of rms voltage 200 V and frequency 50 Hz is applied to a series LR circuit having resistance 20 Ω and inductance 0.1 H. Calculate (i) impedance of the circuit, (ii) rms current, and (iii) phase angle between current and voltage. [A][CO2]

17. Examine the phase relationship between current and voltage in a series LCR circuit under different frequency conditions. [An][CO2]

18. Analyse the performance of a half-wave rectifier with respect to the nature of rectified output, rectification efficiency, and ripple factor. [An][CO3]

19. In a Zener diode, the reverse current increases from 2 mA to 20 mA while the voltage across it increases from 5.1 V to 5.15 V. Calculate the dynamic resistance of the Zener diode in breakdown region. How will the dynamic resistance change if the change in current is 50mA? [A][CO3]

20. A Zener diode with V_Z=6.2V is used to regulate the output voltage from a 12 V DC supply. The load resistance is 1 kΩ and the Zener current required for regulation is 5 mA. Calculate: [A][CO3]
    (a) Series resistance R_S
    (b) Zener current I_Z

21. Explain the phase relationship between input and output in a CE amplifier and its significance. [U][CO4]

22. Compare and analyse inverting and non-inverting amplifiers in terms of gain, phase, and applications. [An][CO5]

[5x6 = 30]